Research overview

Academic career

- Master of Mathematical Physics (MPhys), University of Edinburgh, (2012-2017)
- PhD, Université Paris Cité, *Galaxy cluster cosmology with Euclid*, (2017-2020)
  - With 6 months as a visiting researcher at the California Institute of Technology, USA
- Postdoctoral researcher, LPSC, APC and Cosmostat (2020-Ongoing)

Principle research

- Cosmology with galaxy clusters
  - Parameter inference: Selection functions for Euclid [1], within the Euclid ground-segment first stacked Euclid cluster shear profiles and validation of crowded field photometry, development of new likelihood methods [2,3] and recently simulation based inference approaches [4]
  - Weak gravitational lensing around galaxy clusters: Individual mass estimates [5], new improved cluster mass estimation with magnification [6], contributions to the Euclid ground segment
- Cosmology with cross-correlations: Weak lensing, Galaxy clustering, and CMB lensing
  - f(R) constraints [7], gravito-magnetic lensing [8], dipoles [9,10]

Overview

I am an observational cosmologist and an expert in weak gravitational lensing and galaxy cluster cosmology. I have worked on improving methods for cosmological parameter inference for cluster cosmology, developed new statistical methods to calculate cluster masses from weak gravitational lensing and worked within the Euclid consortium to better understand selection effects for cluster cosmology. My work exemplifies a deep understanding of cluster cosmology, coupled with an ability to develop innovative approaches to unravel the mysteries of dark energy and the large-scale structure of the Universe.

Cluster cosmology

Galaxy clusters are the most massive gravitationally bound objects in the Universe. The study of galaxy clusters has provided evidence of dark matter [15], the primordial origin of density fluctuations in the Universe [4] and the existence of dark energy [14]. Today, counting the abundance of galaxy clusters across cosmic time stands as one of the most important probes in cosmology, as clusters enable us to investigate both the growth of structure and the geometry of the Universe.

Dark energy suppresses the formation of galaxy clusters through accelerating the expansion of the Universe. This is illustrated with N-body simulations in Figure 1, where we clearly see that the matter dominated Universe has more structure and will therefore contain more galaxy clusters. By counting the number of clusters through cosmic time we can constrain the density of dark energy in our Universe.

In order to accurately constrain cosmological parameters using the abundance of galaxy clusters we need to understand precisely their statistical distribution. During my PhD, with the assumption of a Poisson distribution, I showed with Dr. Emmanuel Artis the necessary precision at which the prediction for the mean number of clusters must be known so as not to degrade or bias the cosmological inference [1]. I further developed this work during my postdoctoral research in Grenoble with Dr. Constantin Payerne and Dr. Céline Combet; moving beyond the Poisson assumption, we characterised the accuracy of the usual cluster likelihood methods [11] and subsequently developed a more accurate likelihood which leverages more cosmological information [12].

These previous studies ignored the complexities of actual cluster observations, in particular the relation between dark matter halos and the cluster richness (which is essentially the number of galaxies identified

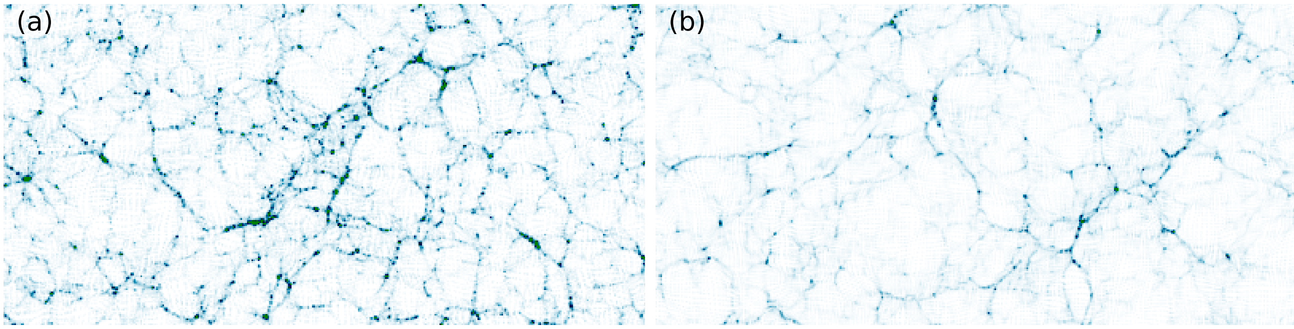

Figure 1: Density fields at a redshift of zero created from the Quijote simulations of a matter dominate universe (a) and a dark-energy dominated universe (b).

within a cluster). When incorporating observational systematics into cluster likelihood analyses the likelihood computation quickly becomes a large-dimensional integral which is computationally expensive to compute. Further to arrive at a tractable expression, the likelihood requires many simplifications, for example log-normal relations between different observable cluster properties. **Since joining CosmoStat I have been developing a more sophisticated method using Simulation Based Inference, which makes it possible to include complex halo-cluster relations and spatially varying observational systematics.**

Within Euclid for the past two years I have been leading a crucial project on selection effects within cluster analyses. By applying the Euclid cluster processing functions to the Euclid Flagship Simulation [2] which provides a mock catalogue of galaxies with Euclid-like observations, we have been able to assess the amplitude of many observational systematics, at a level previously impossible. The large amount of work and attention to detail of this project has led to developments in many parts of the Euclid cluster pipeline. In particular the methods for detecting clusters, measuring the cluster richness and the speed of the cluster weak lensing mass measurement code. This work has led to a better understanding of optical selection effects for galaxy clusters (with an article in preparation [7]) and validation of the Euclid cluster pipeline. The expertise acquired during this project had led me to play an important role in validating the first Euclid cluster catalogue; verifying the photometric cluster properties, photometric redshift estimates and source galaxy selection choices.

Weak lensing for cluster mass estimation

Accurate cosmological constraints from galaxy clusters require precise mass estimates. Weak gravitational lensing provides a robust method for cluster mass estimation that is largely insensitive to the complexities of baryonic physics at small scales. Lensing distorts the statistical properties of background galaxies and can be categorised into two distinct phenomena: weak lensing shear, where galaxy shapes are distorted, and weak lensing magnification, where solid angles on the sky are magnified. As the magnitude of these effects depend on the mass along the line of sight, we can measure cluster masses using gravitational lensing. An example of the expected shear field around a massive cluster is shown in Figure 2.

I have contributed to both areas of study (see also Figure 2). **For weak lensing shear, I produced the largest catalogue of shear-estimated cluster masses [10], using a matched filter method I developed and Hyper Suprime Cam observations.** For weak lensing magnification, I estimated stacked cluster masses [9] using a Wiener filter-based method that significantly reduces noise from galaxy clustering in magnification-based mass estimations and is also applicable to shear-based mass estimation. **This innovative method for reducing noise on cluster mass estimates improved upon what had previously been referred to as the cosmic-variance limit on cluster mass estimation [3].**

Within the Euclid science ground segment, responsible for the processing of Euclid data, I have produced both galaxy cluster catalogues and the first stacked shear profiles around Euclid detected

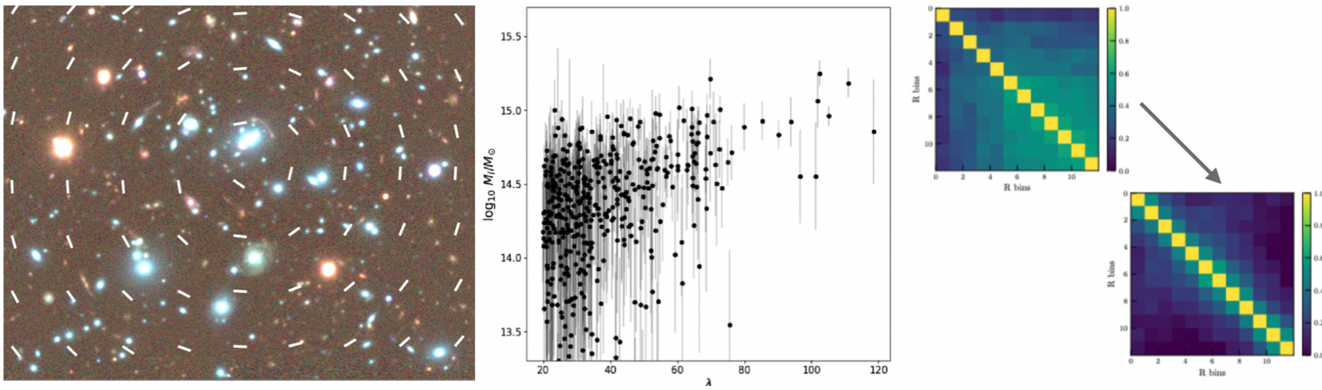

Figure 2: Left: A high mass Hyper Suprime Cam cluster, analysed in [10], the white line segments are an illustration of a tangential shear-field. Middle: the largest catalogue of individual weak lensing masses from [10]. Right: the reduction of off-diagonal noise in the magnification profiles of clusters using a new Wiener filter based method from [9].

galaxy clusters. This work is an important validation step for Euclid clusters pipeline.

Cross-correlations and other projects

Parallel to my work on galaxy clusters, I have explored other topics within cosmology. Here I describe projects which have led to two publications and two articles in preparation.

With Dr. Raphael Kou and Professor James G. Bartlett we used the cross-correlation of Cosmic Microwave Background (CMB) lensing and galaxy clustering to constrain the $f(R)$ model of gravity [5]. Modified gravity theories are a possible manner of explaining the observed accelerated expansion of the Universe. The $f(R)$ model can reproduce ΛCDM expansion history whilst changing the growth of structure, leading to observational differences between the two models. We showed that current observations are in agreement with General Relativity and do therefore not require modifications to gravity. **This work was the first to use the CMB lensing-galaxy cross-correlation to test Hu-Sawicki $f(R)$ gravity.**

Gravitational lensing effects are primarily driven by fluctuations in the cosmic matter density field. However, the motion of massive objects, the cosmic momentum field, also induces gravitational lensing effects. These gravito-magnetic effects are relatively small, on the order of v/c compared to the primary lensing effects, where v is the peculiar velocity of the moving mass and c is the speed of light, making them challenging to detect. To observe this subtle effect, we can reconstruct the cosmic momentum field from the matter density field and cross-correlate it with the observed weak lensing field. This effect should be measurable with Euclid. An observation of this effect will allow us to probe directly the dark matter momentum field and provides a test of Lorentz invariance on cosmological scales. This article will soon be submitted to A&A, a presentation can be found online here [1].

Recently an apparent disagreement has emerged between our velocity with respect to a cosmological rest-frame as observed in the CMB and that observed in the distribution of extragalactic objects (for example [13]), this is known as the cosmological dipole tension. Gravitational lensing can modify the velocity inferred from the distribution of extragalactic objects but does not affect the CMB dipole, I separated these two effects and showed that lensing by local structures cannot resolve this tension [6]. The measurement of angular fluctuations of source counts on large-scales is very difficult, even small changes in the photometric calibration across the sky can lead to significant fluctuations in the density of detected objects. With Dr. Ken Ganga we have been exploring observational systematics that could explain the apparent disagreement, which is the topic of an upcoming paper and has been presented at two conferences [8].

[1] https://indico.cern.ch/event/1166693/contributions/5329644/

Mentoring interns and helping with the tutoring of PhD students has been a rewarding and fruitful aspect of my research. So far I have supervised four internships, each offering unique learning experiences for both myself and the interns. This has allowed me to pursue small research projects into aspects of cosmology that I would otherwise not have the time to pursue. Huy Nguyen Phu recently completed a six month internship, in which he used the apparent proper motion of quasars to estimate our acceleration with respect to the galactic centre. This was a rewarding project in which Huy considerably improved upon the control of systematics with respect to previously published research on this topic. Collaborating closely with PhD students has been integral to my research, demonstrating my role as a supervisor and mentor. I have had the fortune of co-authoring three articles with two different PhD students Dr. Raphel Kou and Dr. Constantin Payerne; for two of these articles the original idea of the project was my own. The skill of supervision is also one that must be learned and practised, and is a skill in which I have enjoyed developing.

Summary

My published research demonstrates the innovative and precise approach that I bring to modern cosmology. The breadth of my work shows me as a researcher who seeks to ask and answer interesting questions. I have contributed to the scientific community not only through my own research, but through supervision of internships and additionally through the organisation of Astronomy on Tap in Paris and the Recontres du Ciel et de l'Espace sought to bring cosmology to the wider public within Paris.

Cosmology is entering an exciting era with the advent of Euclid. The skills and expertise I have developed since the start of my PhD position me to undertake the ambitious research programme detailed in my research proposal and to contribute to cosmological research in France.

References

[1] Artis E., Melin J.-B., Bartlett J. G., Murray C., 2021, Astronomy & Astrophysics, 649, A47
[2] Euclid Collaboration et al., 2024, arXiv e-prints, p. arXiv:2405.13495
[3] Hoekstra H., 2003, , 339, 1155
[4] Kaiser N., 1984, On the spatial correlations of Abell clusters.. Vol. 284, doi:10.1086/184341,
[5] Kou R., Murray C., Bartlett J. G., 2023, arXiv preprint arXiv:2311.09936
[6] Murray C., 2022, Monthly Notices of the Royal Astronomical Society, 510, 3098
[7] Murray C., Bhargava S., 2024, in prep.
[8] Murray C., Ganga K., 2024, in prep.
[9] Murray C., Payerne C., Combet C., M. R., , Submitted to AA
[10] Murray C., Bartlett J. G., Artis E., Melin J.-B., 2022, , 512, 4785
[11] Payerne C., Murray C., Combet C., Doux C., Fumagalli A., Penna-Lima M., 2023, , 520, 6223
[12] Payerne C., Murray C., Combet C., Penna-Lima M., 2024, , 532, 381
[13] Secrest N. J., von Hausegger S., Rameez M., Mohayaee R., Sarkar S., Colin J., 2021, , 908, L51
[14] White S. D. M., Navarro J. F., Evrard A. E., Frenk C. S., 1993, , 366, 429
[15] Zwicky F., 1933, Helvetica Physica Acta, 6, 110